If you declare a variable without any initial value, the memory space allocated to that variable will be occupied with some unknown value. These unknown values are called as **"Junk"** or **"Garbage"** values.

```
#include <iostream>
using namespace std;
int main()
{
        int num1, num2, sum;
        cout << num1 << endl;
        cout << num2 << endl;
        cout << num1 + num2;
}
```

In the above program, some unknown values will be occupied in memory that is allocated for the variables **num1** and **num2** and the statement **cout << num1 + num2;** will display the sum of those unknown junk values.

9.13.2 Initialization of variables

Assigning an initial value to a variable during its declaration is called as "Initialization".

Examples:

int num = 100;

float pi = 3.14;

double price = 231.45;

Here, the variables num, pi, and price have been initialized during the declaration. These initial values can be later changed during the program execution.

Illustration 9.6 C++ Program to find the Curved Surface Area of a cylinder (CSA) (CSA = 2 pi r * h)

```
#include <iostream>
using namespace std;
int main()
{
   float pi = 3.14, radius, height, CSA;
   cout << "\n Curved Surface Area of a cylinder";
   cout << "\n Enter Radius (in cm): ";
   cin >> radius;
   cout << "\n Enter Height (in cm): ";
   cin >> height;
   CSA = (2*pi*radius)*height;
   system("cls");
   cout << "\n Radius: " << radius <<"cm";
   cout << "\n Height: " << height << "cm";
   cout << "\n Curved Surface Area of a Cylinder is " << CSA <<" sq. cm.";
}
```

Output:

```
Curved Surface Area of a cylinder
Enter Radius (in cm): 7
Enter Height (in cm): 20
Radius: 7cm
Height: 20cm
Curved Surface Area of a Cylinder is 879.2 sq. cm.
```

Variables that are of the same type can be initialized in a single statement.

Chapter 9 Page 115-151.indd 140 3/24/2020 9:21:05 AM